

FSAE Design Spec Sheet

2026

Competitors: **Please read the Instructions** Tips (tab below) prior to the completion and submission of this sheet.

Car No.	547
School	Gripton State University

Dimensions	Units						
Overall Dimensions	mm	Length:		Width:		Height:	
Wheelbase & Track	mm	Wheelbase:		Front Track:		Rear Track:	
Center of Gravity Design Height	mm	CG Height:		Confirmed Via:			
Mass without driver	kg	Front:		Rear:		Total:	
Weight Distribution with 68kg driver		% Front:		% Left:			

Suspension Parameters	Units	Front			Rear		
Tire Size, Compound and Make							
Wheels (diameter, width)	inch	Diameter (col D):			Diameter (col G):		
		Width (col E):			Width (col H):		
Wheel material and construction							
Suspension Type							
Suspension design travel	mm	Jounce (col D): Rebound (col E):			Jounce (col G): Rebound (col H):		
Wheel rate (chassis to wheel center)	N/mm						
Roll rate (chassis to wheel center)	Nm/deg						
Sprung mass natural frequency	Hz						
Jounce Damping	% critical		at __ mm/sec:			at __ mm/sec:	
Rebound Damping	% critical		at __ mm/sec:			at __ mm/sec:	
Motion ratio	__:1		Type:			Type:	
Ride Camber (Rate of Camber Change)	deg/m						
Roll Camber	deg/deg						
Static Sum Toe (- out, + in)	deg						
Static camber	deg						
Static camber adjustment method							
Anti dive / Anti Squat	%						
Roll center height above ground, static	mm						
Roll center position at 1g lateral acc	mm	Height (col D):			Height (col G):		
		Lateral (col E):			Lateral (col H):		
Front Caster, Trail, and Scrub Radius		Caster (deg):		Kin Trail (mm):		Scrub Rad (mm)	
Front Kingpin Axis		Inclination (deg):		Offset (mm):			
Static Ackermann	%		Adjustable?				
Suspension Adjustment Methods							
Steer Ratio, C-Factor, Steer Arm Length		Steer Ratio (x:1)		c-factor (mm)		Steer Arm Length	

Brake System / Hub & Axle	Units	Front			Rear		
Rotors							
Master Cylinder							
Calipers							
Brake Pad/Lining Material							
Force and Pressures @ 1g Deceleration		Front Pres. (bar):		Rear Pres. (bar):		Pedal Force (kN)	
Upright Assembly							
Hub Bearings							
Axle type, size, and material							

Ergonomics	Units						
Driver Size Adjustments							
Seat (materials, padding/damping)							
Steering Wheel (dia, construction)		Diameter (mm)		Construction			
Shift Actuator (type, location)							
Clutch Actuator (type, location)							
Instrumentation							
Optional: Driver Safety Systems?							

Electrical	Units						
Power Management / Control							

Wiring / Loom / ECM mounting							
Battery / Charging System							
Grounding							
Driver Assist Systems							
Logging / Telemetry							
Special Sensing Technology							

Frame	Units						
Frame Construction							
Material							
Joining method and material							
Bare frame mass with brackets & paint	kg	Target:		Physical Test:			
Torsional stiffness	N-m/deg	Target:		Simulated:		Physical Test:	
Torsional stiffness validation method							
Impact Attenuator configuration							
Impact Attenuator dimensions	mm	Width:		Height:		Depth:	
Impact Attenuator energy capacity	kJ	Energy:		Method:			

Powertrain	Units						
Manufacturer / Model							
Cylinders & Fuel		Cylinders:			Fuel Type:		
Displacement & Compression		Displacement (cc):			Compression (:1):		
Bore & Stroke	mm	Bore:			Stroke:		
Engine Output		Peak Power (kW)		Peak Torque (Nm)			
Design Speeds	rpm	Max Power:		Max Torque:		80% Torque:	
Induction (natural or forced, intercooled)							
Throttle Body / Mechanism							
Fuel Injection System (manf'r, and type)							
Fuel System Sensors (for fuel mapping)							
Fuel Pressure	bar						
Injector location		xx mm before _____ and directed towards _____					
Intake Plenum		Volume (cc):		Runner length (mm):			
Exhaust Header Configuration		Effective Runner Length (mm):			Variation (mm):		
Exhaust Header Diameters		Primary (mm):		Collector (mm):			
Ignition System							
Ignition Timing							
Oiling System (wet/dry sump, mods)							
Engine Lubricants / Friction Treatment							
Coolant System and Radiator location		_____ mounted xx core _____ radiator , xxxx cfm fan mounted to _____					
Fuel Tank Location, Type						Capacity (L):	
Muffler							
Other significant engine modifications							

Drivetrain	Units						
Drive Type							
Differential System							
Final Drive Ratio	:1						
Vehicle Speed @ max power (design) rpm	kph	1st gear:		2nd gear:		3rd gear:	
Vehicle Speed @ max power (design) rpm	kph	4th gear:		5th gear:		6th gear:	
Half shaft size and material							
Axle Joint type and grease used							

Aerodynamics (if applicable)	Units						
Type / Configuration							
Forces (at 80 kph, $\rho = 1.162 \text{ kg/m}^3$)		Downforce (N):		% Front:		Drag (N):	
Coefficients & Reference Area		Cl:		Ref. Area (m^2):		Cd:	
Notable Features (active, etc)							

Other Information	Units						
Body Work (material, process)							
Optional Information							